

Literature status: the unitary Morita 3-equivalence conjecture

Source arXiv:2404.05193; supporting paper arXiv:2410.05120

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Status: literature-implied answer (full conjecture). The supporting paper proves an enhanced version of the conjectured equivalence and explicitly cites the original source. Forgetting the enhancements gives the exact conjecture. This is a retrieval/status result, not a new mathematical solution produced by the run.

Original conjecture

Giovanni Ferrer’s *Foundations for operator algebraic tricategories*, arXiv:2404.05193 [1], defines the W^* -3-category $\mathbf{UmFC}$ of unitary multifusion categories, unitary bimodule categories, unitary bimodule functors, and bimodule natural transformations. Immediately after Example 3.15, on PDF page 26, it states:

“We conjecture there exists an equivalence $\mathbf{Mod}^\dagger : \mathbf{UmFC} \cong \mathbf{FSSU2C}$, where the latter is the W^* -3-category of finitely semisimple unitary 2-categories.”

The four other extraction hits of the form “we ask that” are coherence requirements inside definitions, not open research questions.

Supporting theorem

Quan Chen, Giovanni Ferrer, Brett Hungar, David Penneys, and Sean Sanford, *Manifestly unitary higher Hilbert spaces*, arXiv:2410.05120 [2], cites arXiv:2404.05193 when using its C^* -3-category foundations. Its Theorem C, stated on PDF page 4 and proved in Section 5, asserts

$$\mathbf{Mod}^\dagger : \mathbf{H}^*\mathbf{mFC} \simeq \mathbf{3Hilb}$$

as a unitary equivalence of C^* -3-categories, preserving unitary adjoints and spherical weights.

This is an enhanced form of the source conjecture. The supporting paper also proves the facts needed to forget the enhancements:

- every unitary multifusion category admits the required unitary dual functor and spherical weight (discussion around its Example 2.9);
- every unitary module category admits a faithful module trace (Corollary 3.13), and finite semisimple bimodule categories are uniquely unitarizable (Corollary 5.8, PDF page 53);
- the forgetful 3-functor $\mathbf{H}^*\mathbf{mFC} \rightarrow \mathbf{mFC}$ is fully faithful (Corollary 5.13, PDF page 54);
- unitarizable finite semisimple 2-categories are uniquely unitarizable (the corollary on PDF page 53), and the forgetful functor $\mathbf{3Hilb} \rightarrow \mathbf{3Vect}$ is fully faithful (Corollary D).

Existence supplies lifts of the objects and 1-morphisms in the original unitary 3-categories; full faithfulness and uniqueness make the result independent, up to unitary equivalence, of the chosen enhancement. Forgetting the unitary-adjoint, weight, and trace decorations in Theorem C therefore gives exactly

$$\mathrm{Mod}^\dagger\colon \mathrm{UmFC} \simeq \mathrm{FSSU2C},$$

which is the conjecture in arXiv:2404.05193.

Scope and provenance

The implication uses the later paper’s enhanced theorem together with its own existence and forgetful corollaries. Because the supporting paper does not restate the undecorated conjecture verbatim, this packet is classified as a literature-implied answer. It is not merely the older nonunitary equivalence between multifusion categories and finite semisimple 2-categories. The supporting paper appeared later, has Ferrer as a coauthor, and explicitly cites arXiv:2404.05193. Exact-title, notation, author, and close-variant searches through 13 August 2026 found this as the decisive later answer. No part of this packet is claimed as a new proof.

References

- [1] G. Ferrer, *Foundations for operator algebraic tricategories*, arXiv:2404.05193 (2024).
- [2] Q. Chen, G. Ferrer, B. Hungar, D. Penneys, and S. Sanford, *Manifestly unitary higher Hilbert spaces*, arXiv:2410.05120 (2024; current version inspected in 2026).